

Unpacking Decline and Dominance: A Machine Learning Framework to Investigate Performance Divergence Between Two Elite Football Clubs

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All the code and csv files can be accessed [here](#)

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Abstract

Manchester United clinched their 20th league title in May 2013, marking the climax of the most successful era in the club's history. Under the legendary management of Sir Alex Ferguson, widely regarded as the greatest manager in British football, they secured 13 Premier League titles, 5 FA Cups, 2 UEFA Champions League trophies, 4 League Cups, and numerous other domestic and international accolades. By 2013, United had firmly established themselves as the dominant force in English football, boasting a global fanbase and a legacy rivaled by few across the globe. In contrast, their cross-town rivals Manchester City had spent much of the previous decades in relative obscurity, often struggling to maintain top-flight status. However, following their acquisition by the Abu Dhabi State and a subsequent influx of investment, City began to reshape their identity. Their dramatic Premier League triumph in 2012—secured on the final day at United's expense—signaled the beginning of a power shift in Manchester football.

Since Ferguson's retirement in 2013, the balance of power has tilted decisively. Manchester City have claimed 7 league titles in 12 seasons, while United have managed only two distant second-place finishes and, more recently, a dramatic decline culminating in a 15th-place finish in the 2024–25 Premier League season. This study explores the divergent trajectories of the two clubs in the post-Ferguson era through a machine learning lens.

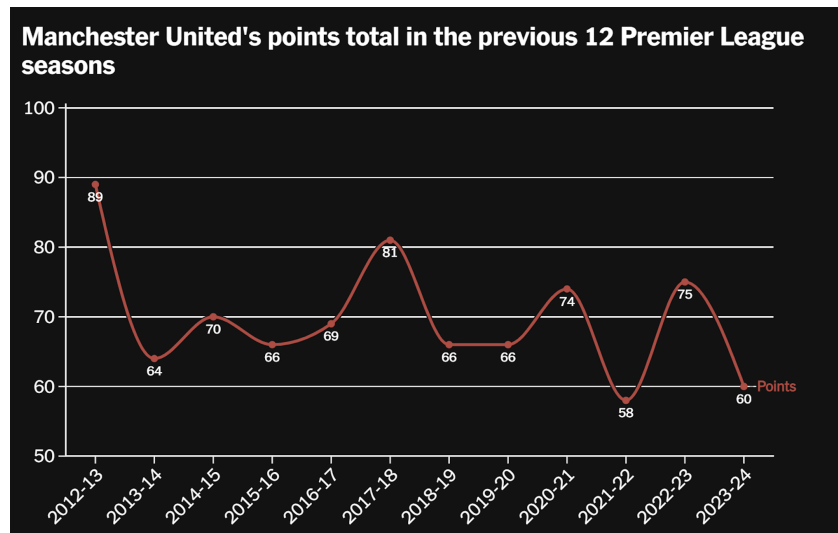


Figure 1: United's decline in points tally across seasons

Introduction

The aims of this study are **two-fold**. **First**, it seeks to examine the underlying rationale behind the stark contrast in league performance between the two Manchester across a six-year period during which both teams experienced multiple managerial changes. **Second**, it aims to explore the extent to which advanced metrics such as **Expected Goals** (xG) and **Possession** can provide deeper insights into the performance shortcomings of each club over that span.

To better structure this investigation, the study has been divided into **two parts**. The first part focuses on the **2013/14 and 2014/15 seasons**, a period preceding the widespread availability of xG data. During this era, performance evaluation is based on objective metrics such as Elo ratings, rest days (days between games for each team and their opponents), pre-match betting odds, and the final league positions of opponents. The second part analyzes the **2015/16 to 2018/19 seasons**, incorporating these same metrics while introducing two additional features, xG_diff (expected goals difference) and **Poss**, to assess how much further explanatory power they add to our understanding of match outcomes.

The Machine Learning technique used in this study is **K-Means Clustering (KMC)**. The objective is to identify the archetypal teams that Manchester United and Manchester City have failed to beat across these six seasons, and to examine whether discernible patterns or contrasts emerge. KMC will be applied in several scenarios (such as dividing datasets up for different managers and applying KMC for each manager) to better evaluate areas where United have consistently struggled and where City have been successful.

Elo-Ratings

Elo ratings, originally developed by Hungarian-American physicist Arpad Elo, were first designed to rank chess players based on relative skill. The system has since been adapted for various competitive domains including football, where it serves as a dynamic measure of team form. In contrast to traditional league standings, which are static and often coarse, Elo ratings provide a probabilistic and continuously updated assessment of performance, making them well-suited for longitudinal analysis in sports analytics.

The mathematical foundation of the Elo system is based on the principle of expected outcomes. When two teams face each other, their current ratings determine the expected result; the winner gains points and the loser loses points, with the magnitude of the rating change depending on the difference in their pre-match ratings and the match outcome. Upsets (where a lower-rated team defeats a higher-rated one) results in a larger rating adjustment. In football applications, additional factors such as goal margin, home advantage, and match importance (e.g., league vs. cup matches) are often incorporated to refine the model's sensitivity. This makes Elo ratings a robust and objective performance metric, particularly useful in pre- xG eras where detailed match statistics were not readily available.

When two teams A and B with ratings R_A and R_B respectively face off, the expected score (or win probability) for team A is given by:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}} \quad (1)$$

Similarly, the expected score for team B is:

$$E_B = \frac{1}{1 + 10^{(R_A - R_B)/400}} = 1 - E_A \quad (2)$$

After the match, the ratings are updated according to the actual outcome. If team A achieves a result S_A (1 for win, 0.5 for draw, 0 for loss), the new rating is calculated as:

$$R'_A = R_A + K(S_A - E_A) \quad (3)$$

where K is the adjustment factor, often called the K-factor, which determines the sensitivity of the rating system to new results. A larger K places more weight on recent matches, while a smaller K leads to more conservative changes.

Expected Goals - xG

Expected Goals (xG) is a widely adopted metric in modern football analytics used to quantify the quality of scoring opportunities. Unlike conventional metrics that simply record whether a shot resulted in a goal, xG assigns a value between 0 and 1 to every shot based on the probability that a similar shot has historically resulted in a goal. For instance, a shot from the edge of the six-yard box might carry an xG value of 0.75, indicating a 75% chance of scoring based on historical data.

These probabilities are derived from machine learning models trained on large datasets of historical shots and their outcomes. The models consider a variety of features including shot distance, angle to goal, type of assist, whether the shot was taken with the foot or head, defensive pressure, and even the phase of play (e.g., open play vs. set-piece). By aggregating xG values across a match or season, analysts can evaluate a team's chance creation performance more objectively than traditional stats like shots or possession.

Mathematically, a common formulation of xG models uses logistic regression, where the likelihood of a goal is estimated as:

$$P(\text{Goal}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}} \quad (4)$$

Here, $x_1, x_2, \dots, x_n$ represent the shot-related features, and $\beta_0, \beta_1, \dots, \beta_n$ are parameters learned from the data. The resulting probability is interpreted as the xG value of the shot. In this project, we will use **xG_{diff}** —the difference between a team's xG and their opponent's xG in a given match—to assess overall performance and determine whether a team underperformed relative to the quality of chances created. This helps differentiate between tactical inefficacy and mere finishing variance when points are dropped.



Figure 2: Example of an xG computation

K-Means Clustering

K-Means Clustering (KMC) is an unsupervised machine learning algorithm designed to partition a dataset into k distinct, non-overlapping clusters. Each cluster is defined by its centroid, which is the mean of all data points assigned to that cluster. The objective of KMC is to minimize intra-cluster variance while maximizing inter-cluster separation.

Algorithmic Steps

The algorithm proceeds through the following iterative process:

1. **Initialization:** Choose k initial centroids randomly from the dataset. These centroids represent the initial cluster centers $C_1, C_2, \dots, C_k$.
2. **Assignment:** For each data point D_i in the dataset D , assign it to the cluster C_j whose centroid is nearest in terms of Euclidean distance:

$$\text{distance}(D_i, C_j) = \sqrt{(x_i - \mu_{j,x})^2 + (y_i - \mu_{j,y})^2}$$

where (x_i, y_i) are the coordinates of data point D_i , and $(\mu_{j,x}, \mu_{j,y})$ is the centroid of cluster C_j .

3. **Update:** After all points have been assigned, update the centroid of each cluster to be the mean of the points assigned to it:

$$\mu_j = \frac{1}{|C_j|} \sum_{D_i \in C_j} D_i$$

4. **Repeat:** Repeat steps 2 and 3 until the centroids no longer change or the changes fall below a predefined threshold.

In contrast to simple averaging methods, KMC provides a more nuanced analysis of patterns in data. Consider the case where a team underperforms on matchdays 1, 2, 3, 4, 35, 36, 37, 38. A naive average of matchday indices suggests failure occurred around matchday 19.5, which misrepresents the actual temporal distribution of failures. Using $k = 2$ for example, KMC would identify two separate clusters — one at the beginning and another at the end of the season which reveals a more accurate picture.

Selecting the appropriate value for k is critical. A poor choice can result in meaningless clusters that obscure important patterns. In this project, we use **Silhouette Analysis** to evaluate the quality of clustering across different values of k .

The silhouette coefficient for a single point i is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where:

- $a(i)$ is the mean intra-cluster distance (i.e., the average distance between i and all other points in the same cluster),
- $b(i)$ is the mean nearest-cluster distance (i.e., the lowest average distance between i and points in any other cluster).

The value $s(i)$ ranges from -1 to 1 , with higher values indicating better-defined clusters. The average silhouette score across all data points is used to determine the optimal k .

In our context, KMC will be used to identify **clusters of matches where Manchester United or Manchester City failed to win**. By grouping such performances based on features like rest days, opponent strength, and match statistics, we aim to extract interpretable patterns that explain underperformance or sustained success.

Model Features

As previously outlined, this study segments the post-Ferguson era into two distinct analytical periods — one without access to Expected Goals (xG) data and one with xG data available. To maintain a balanced and fair comparison, the non- xG era covers the 2013/14 and 2014/15 Premier League seasons, while the xG era spans the four subsequent seasons from 2015/16 to 2018/19. The following features were used in the K-Means Clustering (KMC) model:

- **Matchday**: Indicates the gameweek in which Manchester United or Manchester City faced a particular opponent. Each Premier League season consists of 38 matchdays, typically spanning from August to May, with intermittent breaks for international fixtures.
- **Home/Away**: To numerically encode match venue, a binary variable is used— 0 denotes a home fixture (played at Old Trafford or the Etihad Stadium), and 1 denotes an away fixture (played at the opponent’s ground).
- **Goals For (GF) and Goals Against (GA)**: Represent the number of goals scored and conceded by the team in the match, respectively.

- **Points Won:** A value of 1 indicates a draw and 0 indicates a defeat. Since the focus of this study is on identifying and analyzing failures (i.e., non-wins), all matches that resulted in a win for the club have been excluded to reduce noise and maintain consistency in the clustering process.
- **Elo Difference:** Calculated as the Elo rating of Manchester United/City minus the Elo rating of their opponent on the day of the match.
- **Days Since Last Game:** Measures the rest period for the club between two consecutive league matches. A longer rest may correlate with better performance due to increased preparation and recovery time.
- **xG Difference:** Available only for the xG era, this is calculated as the team’s xG minus their opponent’s xG for a given match. It offers a more granular view of attacking performance and chance creation, independent of actual goals scored.
- **Possession:** Represents the percentage of total match time during which Manchester United or Manchester City controlled the ball. While possession alone does not fully capture attacking intent or dominance across the duration of the game, it serves as a foundational metric to approximate match dynamics. Lower possession values often suggest a tactical preference for defending and counter-attacking, whereas higher possession figures (typically exceeding 50) generally reflect a team’s attempt to dictate the tempo, apply pressure, and generate a higher volume of attacking opportunities.
- **Odds of United/City Win:** This feature serves as a proxy for the many contextual factors surrounding a match that are otherwise difficult to quantify directly. Betting odds implicitly incorporate a wide range of variables such as home advantage, recent form, squad injuries, suspensions, and the strategic incentives for each side to pursue a win. For instance, certain teams may adopt a conservative approach and play for a draw, especially when facing stronger opposition—judging that the marginal gain from a win does not outweigh the risks involved in committing players forward and potentially conceding goals. By relying on these odds, we obtain a reasonably aggregated estimate of pre-match expectations, allowing us to account for the otherwise unquantifiable randomness and nuances that influence match outcomes.

Note: The model will only consider matches in which Manchester United or Manchester City still had a tangible objective to pursue—such as winning the league or securing European qualification. Games played after such objectives were no longer attainable (or had already been secured) are excluded unless they are unlikely to introduce significant noise into the analysis. Season-by-season decisions are as follows:

- **2013/14:** Only the 34 games managed by David Moyes have been included for Manchester United, as the club was no longer in contention for European qualification during the final matchdays. The final four games, overseen by interim manager Ryan Giggs, were excluded. For City, all 38 fixtures are considered, as they clinched the league title on the final matchday.
- **2014/15:** Manchester City secured Champions League qualification by Gameweek 35 and were mathematically out of title contention by Gameweek 36. United secured Champions League football by Gameweek 37. As City won all their post-qualification matches and United dropped points in only one, all 38 matches are included for both clubs to maintain consistency without significantly affecting model integrity.

- **2015/16:** Both clubs were directly competing for the final Champions League spot until the last matchday, finishing level on points. Therefore, all 38 games are included.
- **2016/17:** City secured third place and Champions League qualification on the final matchday, while United were competing to finish sixth. Thus, all fixtures from both teams are relevant and retained.
- **2017/18:** City won the title by Gameweek 34 but were chasing a record 100-point tally, which they achieved on the final day—providing sufficient incentive to consider all 38 matches. United, on the other hand, had secured second place and Champions League football by Gameweek 35. Consequently, their rotated-squad fixtures in Gameweeks 36 and 37 are excluded from the dataset.
- **2018/19:** City and Liverpool were locked in a title race until the final day, warranting inclusion of all City fixtures. United were eliminated from Champions League contention by Gameweek 37, yet their defeat to Cardiff City in Gameweek 38 is included, as they fielded a full-strength squad and the result represents an outlier unlikely to distort the overall analysis.

Methodology

The primary objective of this study is to identify and compare patterns in the types of teams against which Manchester United and Manchester City dropped points over a six-year period. To achieve this, the following steps are undertaken:

- Conduct a silhouette coefficient analysis on all datasets for values of k ranging from 2 to 10, in order to determine the optimal number of clusters for K-Means Clustering (KMC).
- Using the optimal k value obtained, apply KMC to each dataset to identify the centroids of the clusters and, consequently, to determine the average profile of teams that United and City failed to beat during the study period.

The datasets are organised into distinct subsets, each serving a specific analytical purpose:

- **Pre- xG Era (2013/14–2014/15):** Draws and defeats recorded before xG data became readily available, incorporating features such as Elo ratings, rest days, and betting odds.
- **xG Era (2015/16–2018/19):** Draws and defeats recorded during seasons for which xG data is available, allowing the inclusion of xG difference and possession statistics alongside the pre- xG metrics.
- **Manager-Specific Data:** For United, managerial periods are defined as Moyes (2013/14), Van Gaal (2014–2016), and Mourinho (2016–2019). For City, the analysis covers Pellegrini (2013–2016) and Guardiola (2016–2019). Ole Gunnar Solskjær’s interim tenure in the second half of 2018/19 is incorporated into Mourinho’s era for analytical consistency, as both employed a largely similar squad and tactical setup.
- **City’s Title-Winning Seasons:** A focused analysis of City’s championship campaigns during the study period—2013/14 (Pellegrini) in the pre- xG era, and 2017/18 and 2018/19 (Guardiola) in the xG era.

Silhouette Function

To determine the optimal number of clusters (k), we will implement a Silhouette function in Python.

```
1 def silhouette(filename):
2     data = pd.read_csv(filename)
3     # Range of values for k #
4     range_n_clusters = [2, 3, 4, 5, 6, 7, 8, 9, 10]
5     for n_clusters in range_n_clusters:
6         plt.figure()
7         # Silhouette Co-efficient ranges from -1 to 1 #
8         plt.xlim([-1, 1])
9         plt.ylim([0, len(data) + (n_clusters + 1) * 10])
10        # Initialize the clusterer with n_clusters value and a random generator #
11        clusterer = KMeans(n_clusters=n_clusters, random_state=10)
12        cluster_labels = clusterer.fit_predict(data)
13        # The silhouette_score gives the average value for all the samples. #
14        # This gives a perspective into the density/separation of the clusters #
15        silhouette_avg = silhouette_score(data, cluster_labels)
16        print("For # clusters = ", n_clusters,
17              "The average silhouette_score is : ", silhouette_avg)
18        sample_silhouette_values = silhouette_samples(data, cluster_labels)
19        y_lower = 10
20        for i in range(n_clusters):
21            # Aggregate the silhouette scores for samples belonging to
22            # cluster i, and sort them #
23            ith_cluster_silhouette_values = \
24                sample_silhouette_values[cluster_labels == i]
25            ith_cluster_silhouette_values.sort()
26            size_cluster_i = ith_cluster_silhouette_values.shape[0]
27            y_upper = y_lower + size_cluster_i
28            color = cm.nipy_spectral(float(i) / n_clusters)
29            plt.fill_betweenx(np.arange(y_lower, y_upper),
30                             0, ith_cluster_silhouette_values,
31                             facecolor=color, edgecolor=color, alpha=0.7)
32            # Label the silhouette plots with their cluster numbers at the middle #
33            plt.text(-0.15, y_lower + 0.5 * size_cluster_i, str(i))
34            # Compute the new y_lower for next plot #
35            y_lower = y_upper + 10 # 10 for the 0 samples #
36        plt.title("The silhouette plot for the various clusters.")
37        plt.xlabel("The silhouette coefficient values")
38        plt.ylabel("Cluster label")
39        # line for average silhouette score of all the values #
40        plt.axvline(x=silhouette_avg, color="red", linestyle="--")
41        plt.yticks([])
42        plt.xticks([-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1])
43        plt.title(("Silhouette analysis for KMeans clustering on Dataset "
44                  "with n_clusters = %d" % n_clusters),
```



```

45         fontsize=14, fontweight='bold')
46     plt.show()
47     return

```

KMC

Now that we have determined the optimal number of clusters (k), we will compose another function in Python to compute the centroids of the k -clusters.

```

1  from sklearn.cluster import KMeans
2  import pandas as pd
3  import numpy as np
4  def KMC(filename, n):
5      data = pd.read_csv(filename)
6      kmeans = KMeans(n_clusters = n, random_state=0)
7      kmeans.fit(data)
8      centers = kmeans.cluster_centers_
9      # Print features that correspond to the center of the clusters #
10     for c in range(len(centers)):
11         print("-----Values for Cluster", c+1, "-----")
12         for i in range(len(centers[c])):
13             print((data.columns.values[i]), ": ", np.round(centers[c][i],4))
14         print(np.count_nonzero(kmeans.labels_==c),"Games included in this cluster")
15     return

```

Results and Analysis

2013/14 and 2014/15 Seasons

Silhouette analysis for both teams during these two seasons suggests an optimal value of $k = 2$. For united, the silhouette score at that k value is significantly greater than the remaining values but in case of City $k = 3$ is not behind posing a silhouette coefficient only 0.05 smaller.

```

----- Silhouette Values for 2013/14-2014/15 Seasons (United) -----
For # clusters = 2 The average silhouette_score is : 0.6122980205481935
For # clusters = 3 The average silhouette_score is : 0.5357154867770438
For # clusters = 4 The average silhouette_score is : 0.5104549798684642
For # clusters = 5 The average silhouette_score is : 0.4976421730638001
For # clusters = 6 The average silhouette_score is : 0.5153542085482956
For # clusters = 7 The average silhouette_score is : 0.46391344447814675
For # clusters = 8 The average silhouette_score is : 0.4970166890092216
For # clusters = 9 The average silhouette_score is : 0.4840122940705586
For # clusters = 10 The average silhouette_score is : 0.469748347465796

```

Figure 3: United's silhouette scores from 2013-2015

```

----- Silhouette Values for 2013/14-2014/15 Seasons (City) -----
For # clusters = 2 The average silhouette_score is : 0.6859277360671959
For # clusters = 3 The average silhouette_score is : 0.633865327210168
For # clusters = 4 The average silhouette_score is : 0.5630454708194497
For # clusters = 5 The average silhouette_score is : 0.5471912961101704
For # clusters = 6 The average silhouette_score is : 0.5406008053179336
For # clusters = 7 The average silhouette_score is : 0.5564489017214101
For # clusters = 8 The average silhouette_score is : 0.46304394761231504
For # clusters = 9 The average silhouette_score is : 0.4569047010919876
For # clusters = 10 The average silhouette_score is : 0.41402540818363504

```

Figure 4: City's silhouette scores

Running a KMC with $k = 2$ for both teams on this dataset produces:

```

----- Clusters for 2013/14-2014/15 Seasons (United) -----
-----Values for Cluster 1 -----
matchday : 18.0
home_0_away_1 : 0.6154
points_won : 0.5
GF : 0.8462
GA : 1.4615
elo_diff : 189.7388
united_win_odds : 1.8319
opp_final_pos : 11.3077
days_since_last_game_united : 6.8462
days_since_last_game_opp : 7.2308
26 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 21.5455
home_0_away_1 : 0.5455
points_won : 0.3636
GF : 0.3636
GA : 1.7273
elo_diff : -23.8882
united_win_odds : 3.4491
opp_final_pos : 2.3636
days_since_last_game_united : 7.3636
days_since_last_game_opp : 7.0909
11 Games included in this cluster
----- Clusters for 2013/14-2014/15 Seasons (City) -----
-----Values for Cluster 1 -----
matchday : 21.2
home_0_away_1 : 0.7
points_won : 0.4
GF : 1.1
GA : 1.9
elo_diff : 48.086
city_win_odds : 2.564
opp_final_pos : 3.0
days_since_last_game_city : 8.2
days_since_last_game_opp : 8.1
10 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 15.7333
home_0_away_1 : 0.7333
points_won : 0.5333
GF : 1.0
GA : 1.4667
elo_diff : 281.4533
city_win_odds : 1.4593
opp_final_pos : 14.4
days_since_last_game_city : 7.1333
days_since_last_game_opp : 8.1333
15 Games included in this cluster

```

Figure 5: KMC Results for both United and City

The K-Means clustering results for Manchester United during the 2013/14–2014/15 seasons reveal two distinct clusters representing the types of opponents United struggled against in this period.

United’s Centroids: This cluster is characterized by an average matchday of approximately 18, with a home/away value averaging around 0.6, indicating that United struggled more often in away games. The points-won rate is approximately 0.5, reflecting a roughly equal distribution of draws and losses. While United’s goals scored (GF) average of 0.85 suggests they were able to find the net in several matches, the pragmatic tactical setups employed by both Moyes and Van Gaal limited their goal-scoring volume. Conversely, these cautious approaches failed to prevent opponents from scoring, as reflected by a goals conceded (GA) average of about 1.5, indicating an alarming frequency of goals allowed. The positive average Elo difference (roughly 190) shows that United generally faced slightly weaker opponents in this cluster. With winning odds around 1.83 and opponents’ average league position near mid-table (approximately 11th), this cluster comprises of 26 matches where United were expected to perform better but still dropped points against teams likely finishing between 7th and 14th place.

With an average matchday of approximately 21.5, this cluster features a roughly equal distribution of home and away games, with slightly more fixtures occurring in the second half of the season (after matchday 19). The points-won metric (0.36) is lower, indicating a higher proportion of losses compared to draws. The negative Elo difference (-23.9) and the opponents’ average final league position of about 2.4 suggest these sides were title contenders with superior quality and tactical sophistication compared to United, explaining their dominance in these matches. United maintained a pragmatic setup, aiming to counterattack or secure draws, as reflected by a low goals scored (GF) average of 0.36, highlighting their difficulty in scoring. Despite these tactical efforts under Moyes and Van Gaal, United’s opponents, likely buoyed by confidence and championship ambitions, were able to break down their defense, with United conceding nearly 2 goals per game on average. Additionally, United’s relatively high winning odds indicate a marked decline in performance, with finishes of 7th under Moyes and 4th under Van Gaal — a stark contrast to Ferguson’s era, where United were typically favored against title contenders and were championship contenders themselves. This shift in bookmaker expectations further underscores United’s diminished status during this period.

City’s Centroids: Both of City’s clusters suggest that the majority of their dropped points came away from home, with the home-away feature returning average values of ≥ 0.7 in both cases. The first cluster reflects their struggles against fellow title contenders (e.g., Liverpool and Chelsea), while the second cluster offers a more damning explanation for their distant second-place finish behind champions Chelsea in the 2014/15 season. Despite enjoying a substantial positive Elo difference over these opponents, City still dropped points to sides in the bottom half of the table, as indicated by the opponents’ average final league position of 14.4. A possible rationale for this could be that these lower-ranked teams, often without European football and already eliminated from domestic cups, had the advantage of longer rest periods between games (8.13 days on average). City, by contrast, were simultaneously competing on multiple fronts (the Premier League, domestic cups, and the Champions League) which likely forced squad rotation and increased injury risk. It is important to note that the model only accounts for rest days between Premier League fixtures and does not factor in midweek commitments such as FA Cup, League Cup, or Champions League matches, which would have further compressed City’s schedule.

xG Era: 2015/16-2018/19

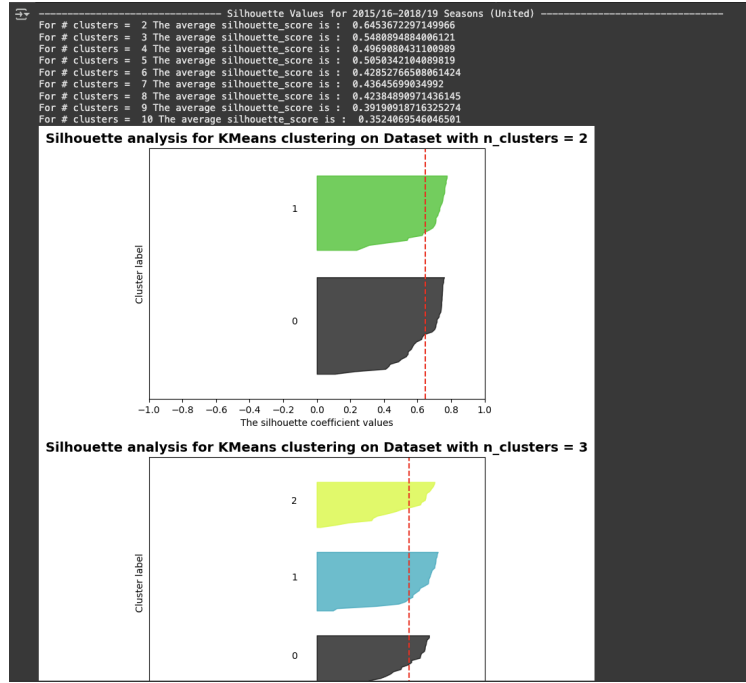


Figure 6: United's silhouette scores and cluster for $k = 2$ from 2015-2019

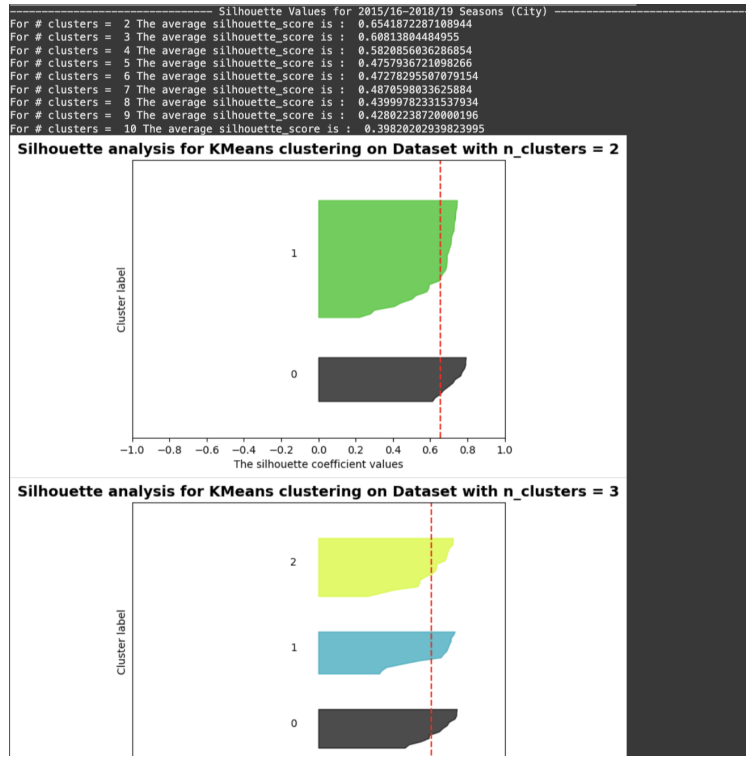


Figure 7: City's silhouette scores and clusters for $k = 2$ and $k = 3$ from 2015-2019

Once again, $k = 2$ yields the highest silhouette coefficient, with $k = 3$ producing a value that is relatively close, similar to what we observed in the pre-xG era scenario. However, for $k = 3$, one cluster in each dataset (C2 for United and C0 for City) contains points that generate negative silhouette scores, which is not the case when $k = 2$. Additionally, for $k = 3$, the points gap between clusters C0, C1, and C2 is approximately 12 points — the same as for $k = 2$. For these reasons, we again select $k = 2$ as the optimal choice for our KMC model in this scenario.

(i) United

With $k = 2$ and two additional features in the model to document *xG difference* and *Possession*, the KMC produces the following results:

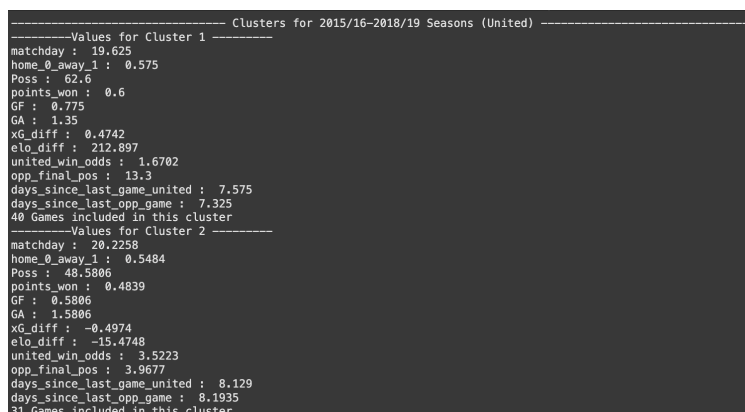


Figure 8: Centroid values for United in the *xG* era

The first cluster highlights a recurring issue faced by Manchester United under both Louis van Gaal and José Mourinho. Both home and away, United consistently struggled to create a high volume of quality chances, often failing to break down mid-table and bottom-half sides (average opponent league position: 13.3). On average, they enjoyed 62.6% possession, meaning the majority of the game was spent in attacking phases. However, they scored less than a goal per game in this cluster while conceding an average of 1.35 goals per match, often from counter-attacks or set pieces. The xG difference, barely amassing to 0.5 despite such a dominant share of possession, is indicative of their inability to create clear-cut opportunities. Conversely, their opponents were notably more clinical, finding the net with a lower xG. This statistical pattern is common in matches where the favoured side dominates possession, creates numerous half-chances but few high-quality opportunities, while the underdog defends in a compact shape and capitalises on transitional moments or set-piece situations.

The second cluster highlights the clear gulf in quality between United and their top-four rivals during these four seasons. United were consistently in contention for Champions League qualification but fell short in three of the four campaigns, finishing 5th (2016), 6th (2017), 2nd (2018), and 6th (2019). A major factor in these shortcomings was their inability to beat fellow top-four contenders, reflected in an average opponent league position of 3.97. By this stage of the post-Ferguson decline, the betting market had lost faith in United’s dominance (average win odds of 3.52), and results justified such scepticism. Against stronger opposition, their attacking struggles intensified — both GF and xG difference were lower than in the first cluster — while defensive frailties were exposed, leading to a higher goals-against average. Although United occasionally sought to control

possession (averaging close to 50%), they were neither as threatening in attack nor as compact defensively as their opponents. This performance gap was compounded by managerial changes, with each coach reshaping the squad to suit their philosophy, in contrast to rivals such as City, Chelsea, and Tottenham, who benefited from more settled squads built around the tactical visions of their managers.

(ii) City

Similarly, for Manchester City, the KMC function with $k = 2$ produces the following cluster centroids:

```

-----Clusters for 2015/16-2018/19 Seasons (City) -----
-----Values for Cluster 1 -----
matchday : 20.8182
home_0_away_1 : 0.5455
Poss : 59.2727
points_won : 0.4545
GF : 0.8788
GA : 1.9091
xG_diff : 0.2439
elo_diff : 92.427
city_win_odds : 2.0788
opp_final_pos : 5.2727
days_since_last_game_city : 8.303
days_since_last_opp_game : 7.5152
33 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 20.6923
home_0_away_1 : 0.6923
Poss : 69.6154
points_won : 0.6923
GF : 0.8462
GA : 1.1538
xG_diff : 0.8562
elo_diff : 321.3308
city_win_odds : 1.3054
opp_final_pos : 13.6154
days_since_last_game_city : 5.7692
days_since_last_opp_game : 6.0769
13 Games included in this cluster

```

Figure 9: Centroid values for City in the xG era

City’s consistency across this four-year period is reflected in the fact that only 46 games are included across both clusters, compared to 71 for United. In each cluster, City dominated possession — averaging 59.27% in Cluster 1 and 69.62% in Cluster 2 — yet struggled to convert this control into goals, scoring less than one goal per game on average (0.8788 and 0.8462, respectively).

Cluster 1 largely represents matches against stronger opposition (average opponent position 5.27) where defensive vulnerabilities were exposed, conceding nearly 1.91 goals per game. These results align closely with Pellegrini’s final season (2015/16) and Guardiola’s transitional first season (2016/17), where City’s expansive pressing was often bypassed by elite sides in transition. The xG difference in this cluster is only 0.2439, indicating that despite dominating the ball, City were creating only marginally better chances than their opponents — a reflection of their lack of attacking cohesion at the time.

Cluster 2, by contrast, represents dropped points against lower-half opposition (average position 13.62) and is more reflective of Guardiola’s later seasons. Here, the xG difference rises sharply to 0.8562, showing that even in the few matches City failed to win during the 2017/18 and 2018/19 campaigns, they were still creating significantly higher-quality chances than their opponents, only falling short in terms of finishing. This statistical shift is a clear indicator of how well City had clicked under Pep from 2017/18 onwards, culminating in their record-breaking 100-point season in 2017/18 and 98-point season in 2018/19. The higher average Elo difference (321.33) in this cluster further highlights the gulf in quality between City and these opponents, making these dropped points statistical outliers in otherwise dominant campaigns.

Manager Specific Breakdown

This section of the study will again employ Silhouette analysis and KMC to inspect manager-by-manager clusters for both City and United. For City, we will analyse Pellegrini's era both with xG as a metric (2015/16) and without xG (2013/14–2014/15), followed by three seasons under Guardiola (2016/17–2018/19). For United, the analysis will cover the 2013/14 season under David Moyes and the 2014/15 season under Louis Van Gaal in the pre- xG era, followed by the 2015/16 season under Van Gaal and the 2016/17–2018/19 seasons under Jose Mourinho, incorporating the additional features of xG and *possession* into the model. Given the limited sample size from Ole Gunnar Solskjær's interim tenure, the analysis will primarily focus on cluster data from José Mourinho's period, with references to Solskjær only where the data aligns with qualitative observations ("eye test"). While both managers employed a similar formation, their in-game philosophies exhibited subtle but notable differences.

Moyes, Louis Van Gaal and Jose Mourinho

As expected, the dataset for each manager produces the best silhouette score for $k = 2$. This is because it is the same data as before, but segmented differently across managerial eras.

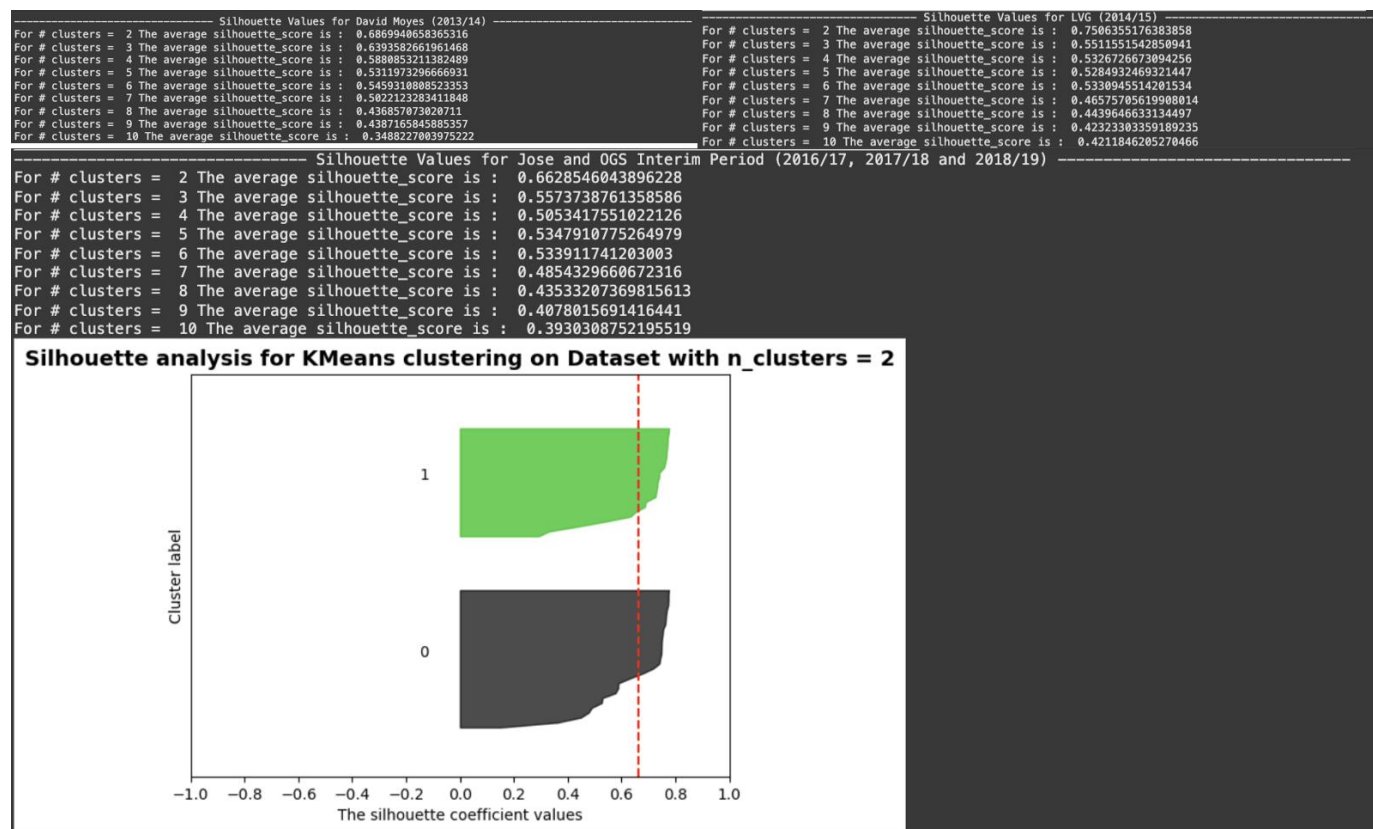


Figure 10: Silhouette scores and Clusters for different United managers

Running KMC on all these datasets with an optimal value of $k = 2$ we get the following cluster centroids:

```

----- Clusters for David Moyes (2013/14) -----
-----Values for Cluster 1 -----
matchday : 18.0
home_0_away_1 : 0.5455
points_won : 0.2727
GF : 0.4545
GA : 1.9091
elo_diff : 43.5727
united_win_odds : 2.9991
opp_final_pos : 3.4545
days_since_last_game_united : 6.7273
days_since_last_game_opp : 6.5455
11 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 15.0
home_0_away_1 : 0.3333
points_won : 0.5
GF : 1.1667
GA : 1.6667
elo_diff : 261.925
united_win_odds : 1.47
opp_final_pos : 13.8333
days_since_last_game_united : 8.1667
days_since_last_game_opp : 8.1667
6 Games included in this cluster

```

Figure 11: Centroid values for United under Moyes 13/14

```

----- Clusters for LVG (2014/15) -----
-----Values for Cluster 1 -----
matchday : 18.0714
home_0_away_1 : 0.7857
points_won : 0.5714
GF : 0.7857
GA : 1.4286
elo_diff : 180.745
united_win_odds : 1.8293
opp_final_pos : 12.1429
days_since_last_game_united : 6.9286
days_since_last_game_opp : 7.3571
14 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 22.25
home_0_away_1 : 0.5
points_won : 0.5
GF : 0.5
GA : 1.0
elo_diff : -78.315
united_win_odds : 3.4975
opp_final_pos : 1.75
days_since_last_game_united : 6.75
days_since_last_game_opp : 7.0
4 Games included in this cluster
----- Clusters for LVG with xG (2015/16) -----
-----Values for Cluster 1 -----
matchday : 23.625
home_0_away_1 : 0.5
Poss : 59.625
points_won : 0.5
GF : 0.5
GA : 1.5
xG_diff : -0.4675
elo_diff : 0.0687
united_win_odds : 2.92
opp_final_pos : 5.375
days_since_last_game_united : 6.25
days_since_last_game_opp : 6.25
8 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 15.8182
home_0_away_1 : 0.7273
Poss : 61.0
points_won : 0.4545
GF : 0.7273
GA : 1.3636
xG_diff : 0.2327
elo_diff : 176.3891
united_win_odds : 1.9227
opp_final_pos : 13.2727
days_since_last_game_united : 7.0
days_since_last_game_opp : 7.0
11 Games included in this cluster

```

Figure 12: Centroid values for United under LVG 2014/15 and 2015/16


```

----- Clusters for Jose and OGS Interim Period (2016/17, 2017/18 and 2018/19) -----
-----Values for Cluster 1 -----
matchday : 19.0435
home_away_1 : 0.5652
Poss : 44.7391
points_won : 0.4783
GF : 0.6087
GA : 1.6087
xG_diff : -0.5078
elo_diff : -20.8813
united_win_odds : 3.7317
opp_final_pos : 3.4783
days_since_last_game_united : 8.7826
days_since_last_opp_game : 8.8696
23 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 21.069
home_away_1 : 0.5172
Poss : 63.2069
points_won : 0.6552
GF : 0.7931
GA : 1.3448
xG_diff : 0.5659
elo_diff : 226.7448
united_win_odds : 1.5745
opp_final_pos : 13.3103
days_since_last_game_united : 7.7931
days_since_last_opp_game : 7.4483
29 Games included in this cluster

```

Figure 13: Centroid values for United under Jose 2016/17-2018/19 and OGS interim period

David Moyes: Reasons for United’s disastrous season under Moyes can be inferred from the average feature values of the two clusters. The first cluster highlights the ease with which top sides (average league position of 3.4545) dominated United, scoring at an alarming rate of almost two goals per game ($GA = 1.9091$) while largely restricting United’s attacking output ($GF = 0.4545$). While such struggles against elite opposition might have been expected given the magnitude of replacing Sir Alex Ferguson, the second cluster reveals an abnormally high number of points dropped at home to bottom-half opposition (average opponent final position ≈ 14 , $home/away = 0.333$). United’s defensive frailties meant that even in matches where they found the net, they were unable to contain teams that typically struggled to score, hence their lower league finishes.

Both clusters underscore two key issues:

- **Squad quality and recruitment:** United failed to bring in reinforcements, while rivals such as City, Chelsea, and Liverpool strengthened considerably. Moyes was left with an aging squad that had arguably overperformed in 2012/13. The lack of fresh talent in both defence and attack contributed to subpar performances in both **GF** and **GA** metrics.
- **Tactical misalignment:** Under Ferguson, United employed a high-pressing, transition-focused style that was ruthless in counter-attacking phases. Moyes introduced a reactive approach, relying heavily on deep defensive blocks and aerial deliveries via crosses and long balls. This ideological shift, applied to an aging squad, with no new incomings suited to the system’s demands, inevitably produced sub-optimal results.

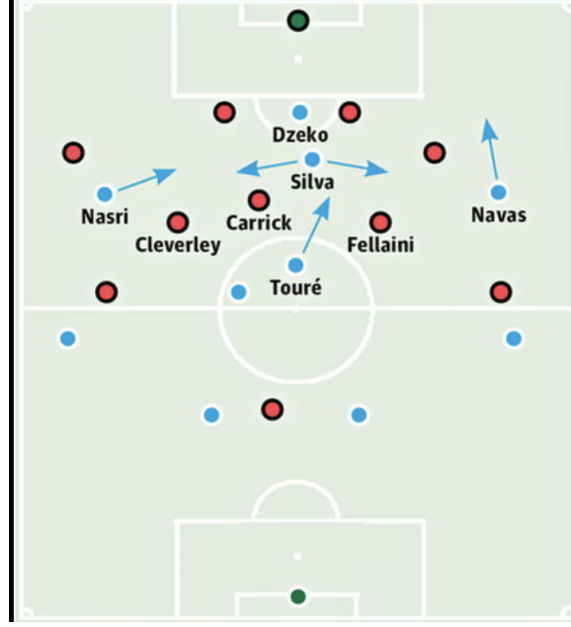


Figure 14: Diagrammatic illustration of the tactical flaws in Moyes’ reactive approach against top sides (City)

Louis Van Gaal: The two different cluster sets of data (2014/15 and 2015/16) provide a useful comparison point for the additional context that *possession* and *xG_difference* offer for our analysis. United finished 4th in 2014/15 and 5th in 2015/16, with similar points tallies (70 and 66 respectively). For 2014/15, the data shows that United actually performed well against stronger sides, with only four games included in the cluster featuring opponents with an average *opp_final_pos* of 1.75, and conceding just one goal per game on average. Their main struggles that season came against mid-table and lower-half teams, as evidenced by an average *topp_final_pos* of 12.1429. A majority of these matches were away from home (*home_0_away* = 0.7857).

This pattern may be explained by the club’s transfer strategy. After finishing 7th under Moyes, United spent over £200 million on both established and emerging players in Europe. Consequently, Van Gaal had to integrate a large number of new signings into his squad while implementing a system that represented a significant tactical overhaul compared to Moyes’ approach. These changes created challenges in execution and consistency. Nevertheless, the influx of new players and the benefit of additional rest due to the absence of European football were sufficient for United to secure a return to the Champions League with a 4th place finish.

The second set of clusters provides more detailed insight as to why United struggled to push-on from their 4th place finish in 14/15 and challenge for the title in 15/16 stagnating instead. In both clusters *Poss* = 59.625 and 61.0 respectively meaning United often dominated the ball against their opponents both higher up in the table. In both clusters, *Poss* = 59.625 and 61.0 respectively, meaning United often dominated the ball against their opponents both higher up in the table (*opp_final_pos* = 5.375) and in the bottom half (*opp_final_pos* = 13.2727). Sides higher up in the table were able to cut off Van Gaal’s predictable passing patterns in the final third and bypass his system’s counter-press to create better quality opportunities with less position, this can be seen with *GA*=1.5 *GF*=0.5 and *xG_diff* = 0.4675, a pattern observed in equal density home and away.

United also dominated possession against sides that, on average, finished in the bottom half of the table, yet failed to create high-quality goal-scoring opportunities in greater volume than their opponents. Despite enjoying more than 60% possession, they could only produce an xG_{diff} of 0.2327, which is close to negligible. Their opponents, often spurred on by their home crowd ($home_0_away_1 = 0.7273$), created chances in lower volume but of higher quality, while remaining compact defensively and scoring more goals on average ($GA = 1.3636 > GF = 0.7273$). This lack of quality chance creation and goal threat can be attributed to the departure of Robin van Persie and Ángel Di María prior to the season, coupled with the reliance on young attackers such as Anthony Martial and Marcus Rashford for much of the campaign due to injuries. Additionally, attackers were often instructed to avoid shooting first time, instead being encouraged to take extra touches in pursuit of creating a higher-quality chance. While well-intentioned, this approach significantly reduced United's shot volume and limited opportunities to test the goalkeeper, thereby contributing to a consistently low xG regardless of the quality of the opposition. The tactical system lacked incisiveness, and there was not enough individual quality to decisively influence games, as evidenced by the cluster data. That season, United averaged the league's second-highest possession but ranked among the bottom three for chance creation.



Manchester Utd		
Premier League this season		
		PL rank
Goals	46	11th
Total shots	418	16th
Shots on target	139	15th
Chances created	301	18th
Passes into final 3rd	2327	14th
Passes backwards	3,107	1st
Sideways passes	9,718	3rd
Average possession	58.23%	2nd

Louis Van Gaal

Figure 15: Graphic to illustrate United's struggles with chance creation and Goalscoring: Sky

Jose Mourinho and OGS Interim: José Mourinho's United would set up based on the quality of their opponents while retaining a largely pragmatic defensive structure. Against stronger teams, both home and away, they would adopt a compact shape with less emphasis on pressing, focusing instead on direct play to exploit open spaces using quick attackers such as Rashford, Martial, Lingard, and Lukaku, with Paul Pogba tasked with delivering incisive forward passes. However, these stronger sides were generally well-equipped to neutralise such direct attacks in transition or to deal with United's long balls, and their own attacking quality meant that sustained pressure often resulted in goals. A negative xG_{diff} , possession $< 50\%$, and an average $opp_final_pos = 3.4783$ largely reflect these game dynamics, with United relying heavily on top-quality defenders to ensure Mourinho's approach yielded consistent results. While reinforcements such as Lindelöf and Eric Bailly were brought in to bolster the backline, neither developed into the world-class calibre central defenders that are essential for a Mourinho side to succeed, as evidenced in his previous teams: Chelsea (Terry), Porto (Ricardo Carvalho), Inter (Zanetti), and Real Madrid (Ramos and Pepe).

The second cluster builds on this pattern but highlights matches against bottom-half and mid-table opposition, where United enjoyed the majority of possession and created a greater number of chances ($xG_{diff} = 0.5659$) but struggled with finishing. This was also observed during Ole Gunnar Solskjær’s interim spell, where, after an initial burst of form from Rashford and Martial, consistent goals became scarce, particularly as Lukaku, United’s main goalscoring outlet, experienced a notable dip in form. Potent attacking threats have historically been a hallmark of successful Mourinho sides, such as Drogba and Lampard (Chelsea), Eto’o and Milito (Inter), and Ronaldo, Benzema, and Di María (Real Madrid). One of Mourinho’s main grievances, alongside the lack of elite defensive reinforcements, was the club’s failure to provide experienced attacking talent to complement the younger options in Rashford and Martial, who, as developing players, could not be relied upon week in and week out, and Lukaku, whose goals tended to come in bursts rather than consistently throughout the season — a recurring trend in both 2017/18 and 2018/19.

These two key interpretations of the cluster data help explain why, despite winning two trophies in his first season and finishing second with 80+ points in his second season (the highest total by any United side since Sir Alex Ferguson’s retirement), Mourinho was ultimately unable to restore United to the summit of English football.



Figure 16: An illustration of how the striker and full-backs are utilised for direct attacking play in transition under Mourinho

Manuel Pellegrini and Pep Guardiola

The *xG*-era silhouette for Pellegrini (2015-16) returned the highest silhouette score for $k = 3$, first instance in this study so far. The pre-xG era for Pellegrini and for the three seasons under Guardiola all returned highest silhouette co-efficients for $k = 2$. To avoid repetitiveness in analysis, we will present the title-winning seasons for both managers as sub-sections within their respective era-wise inspections.

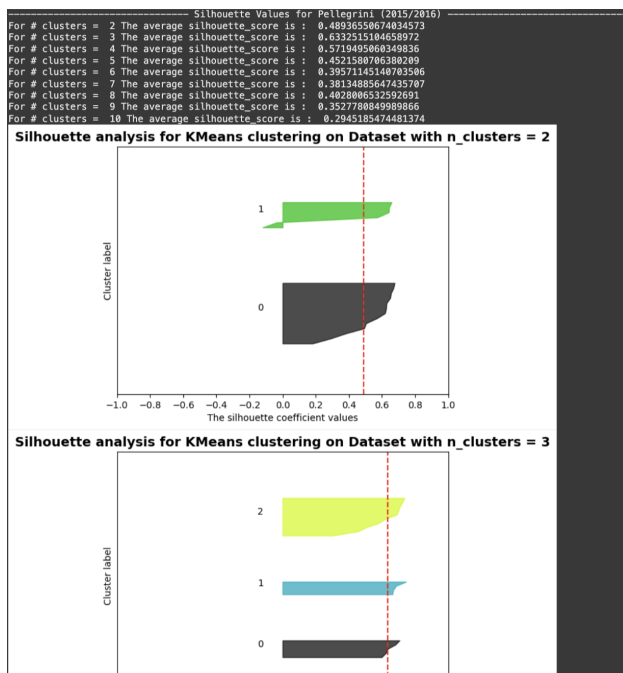


Figure 17: Silhouette scores and cluster panel for Pellegrini in 15/16 with $k = 3$

Running the KMC function on these datasets with $k = 2$ and $k = 3$ for the respective cases we get the following cluster centroid values:

```
-----Values for Cluster 1 ----- Clusters for Pellegrini (2013/14 and 2014/15) -----
matchday : 21.2
home_0_away_1 : 0.7
points_won : 0.4
GF : 1.1
GA : 1.9
elo_diff : 48.086
city_win_odds : 2.564
opp_final_pos : 3.0
days_since_last_game_city : 8.2
days_since_last_game_opp : 8.1
10 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 15.7333
home_0_away_1 : 0.7333
points_won : 0.5333
GF : 1.0
GA : 1.4667
elo_diff : 281.4533
city_win_odds : 1.4593
opp_final_pos : 14.4
days_since_last_game_city : 7.1333
days_since_last_game_opp : 8.1333
15 Games included in this cluster
```

Figure 18: Centroid values for Pellegrini in 13/14 and 14/15 with $k = 2$

```

----- Clusters for Pellegrini (2015/2016) -----
-----Values for Cluster 1 -----
matchday : 22.5
home_0_away_1 : 0.7
Poss : 56.1
points_won : 0.4
GF : 0.8
GA : 2.3
xG_diff : -0.27
elo_diff : 153.432
city_win_odds : 1.915
opp_final_pos : 6.6
days_since_last_game_city : 8.4
days_since_last_opp_game : 8.1
10 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 18.75
home_0_away_1 : 0.75
Poss : 63.5
points_won : 0.75
GF : 0.5
GA : 0.75
xG_diff : 0.9525
elo_diff : 296.5425
city_win_odds : 1.475
opp_final_pos : 16.0
days_since_last_game_city : 6.25
days_since_last_opp_game : 6.25
4 Games included in this cluster
-----Values for Cluster 3 -----
matchday : 24.2
home_0_away_1 : 0.4
Poss : 52.4
points_won : 0.4
GF : 0.8
GA : 1.4
xG_diff : 0.112
elo_diff : 48.678
city_win_odds : 2.446
opp_final_pos : 3.4
days_since_last_game_city : 8.0
days_since_last_opp_game : 6.6
5 Games included in this cluster

```

Figure 19: Centroid values for Pellegrini in 15/16 with $k = 3$

```

----- Clusters for Pep (2016/17, 2017/18 and 2018/19) -----
-----Values for Cluster 1 -----
matchday : 21.5556
home_0_away_1 : 0.6667
Poss : 72.3333
points_won : 0.6667
GF : 1.0
GA : 1.3333
xG_diff : 0.8133
elo_diff : 332.3478
city_win_odds : 1.23
opp_final_pos : 12.5556
days_since_last_game_city : 5.5556
days_since_last_opp_game : 6.0
9 Games included in this cluster
-----Values for Cluster 2 -----
matchday : 18.9444
home_0_away_1 : 0.5
Poss : 62.9444
points_won : 0.5
GF : 0.9444
GA : 1.8333
xG_diff : 0.5661
elo_diff : 70.6878
city_win_odds : 2.0678
opp_final_pos : 5.0556
days_since_last_game_city : 8.3333
days_since_last_opp_game : 7.4444
18 Games included in this cluster

```

Figure 20: Centroid values for Pep from 16/17 to 18/19 $k = 2$

Pellegrini: In the clusters for 2013/14 and 2014/15, City’s defensive vulnerabilities stand out as the main factor behind their dropped points. Conceding an average of 1.9 and 1.5 goals across C0 and C1 respectively, against opponents across the table, provides statistical backing for this claim. This was likely linked to the club’s struggle to find a reliable partner for Kompany, with Demichelis and Mangala both failing to meet expectations. Despite these issues, City remained competitive in the league due to the attacking output of Sergio Agüero and Edin Džeko. In the clusters from their title-winning 2013/14 season, points were only dropped in matches where the attack could not compensate for defensive lapses. Across all three clusters for that season, GA averaged 1.75, 1.5, and 1.33 respectively, while GF remained constant at 1.0. The second and third clusters further indicate that City were more likely to drop points when facing opponents with greater rest, as previously noted in the pre- and post- xG era comparisons. In both cases, a notable rest-day advantage for opponents likely forced City into rotation, making them more vulnerable to lower-half

teams. While City possessed superior quality to most sides in these two seasons, occasional slip-ups occurred due to squad weaknesses and fixture congestion.

The 2015/16 cluster values for Pellegrini, in which City finished fourth, reveal a notable trend. City’s xG output was higher against opponents with an average final position of 3.4 (Top-4 contenders) than against those averaging 6.6 (5th–8th). This reflects a shift in game dynamics. Against stronger sides, City averaged 52.4% possession, producing open, end-to-end matches that marginally favoured their xG returns. Against slightly lower-ranked teams, possession rose to 56%, with opponents deliberately conceding territory to disrupt buildup and exploit transitions. This reduced City’s shot volume but increased the danger of chances conceded, echoing patterns observed with Manchester United. C2 is less significant, covering four matches where City, despite clear superiority in possession and chance creation, failed to convert against teams they were expected to beat. This also highlights the impact of Agüero’s absence, as he featured in only 30 of 38 league matches yet still scored 24 goals. The team’s xG underperformance correlates with his limited availability, with neither Negredo nor Bony providing an adequate clinical replacement.

Pep Guardiola: The sustained dominance of Manchester City under Pep Guardiola is evident from the fact that they dropped points in only 27 of 114 league matches across three seasons. When era-based centroids are compared with those from the title-winning campaigns of 2017/18 and 2018/19, it emerges that 15 of these 27 instances occurred in 2016/17, Guardiola’s first full season, in which City finished third. In the title-winning clusters, City continued to dominate both possession (73% and 61%) and xG difference (0.44 and 0.58). These isolated point drops largely reflected matches where City failed to convert chances, while opponents—typically fellow title contenders with an average opponent final position of 3.8 in C2—were more efficient with their opportunities. Both clusters had home/away values ≥ 0.5 , indicating that away-ground conditions likely contributed to reduced attacking conversion and heightened opponent efficiency.

Examining the centroids across Guardiola’s first three seasons (2016/17–2018/19) further explains his initial challenges in England. In 2016/17, the majority of difficulties arose against teams outside the top four: C1 recorded an average opponent league position of 12.56, while C2 averaged 5.06. This suggests that compact, low-block defensive setups—executed effectively by sides such as Everton (4–0, 2016/17) and Chelsea (3–1, 2016/17)—were more problematic than higher-ranked sides, who attempted to match City’s intensity and thus conceded space for City’s passing structures to exploit. By his second season, Guardiola had refined the system to break down low-blocks more consistently, aided by targeted additions in midfield and attack with profiles better suited to executing his positional play principles.



Figure 21: Base structure of Pep’s system in possession


```

----- Silhouette Values for Pep (2017/18 and 2018/19) -----
For # clusters = 2 The average silhouette_score is : 0.7821019262890888
For # clusters = 3 The average silhouette_score is : 0.6050487420827153
For # clusters = 4 The average silhouette_score is : 0.5090407802313724
For # clusters = 5 The average silhouette_score is : 0.527956825714362
For # clusters = 6 The average silhouette_score is : 0.5582039239855192
For # clusters = 7 The average silhouette_score is : 0.48072149955590726
For # clusters = 8 The average silhouette_score is : 0.4233596594163022
For # clusters = 9 The average silhouette_score is : 0.3323498942433129
For # clusters = 10 The average silhouette_score is : 0.16095666205767384

```

Figure 22: Cluster data from 13/14 Title winning season under Pellegrini

```

----- Silhouette Values for Pellegrini (2013/14) -----
For # clusters = 2 The average silhouette_score is : 0.702831800121393
For # clusters = 3 The average silhouette_score is : 0.7160350803901365
For # clusters = 4 The average silhouette_score is : 0.6577622541095512
For # clusters = 5 The average silhouette_score is : 0.5133101564096966
For # clusters = 6 The average silhouette_score is : 0.48831893002324056
For # clusters = 7 The average silhouette_score is : 0.36286949828996184
For # clusters = 8 The average silhouette_score is : 0.29263491423573745
For # clusters = 9 The average silhouette_score is : 0.08875851710151376
For # clusters = 10 The average silhouette_score is : 0.08160161275979229

```

Figure 23: Cluster data for 17/18 and 18/19 title winning seasons under Pep

Conclusion

Where United declined and How City Dominated

- **Constant Manager and Ideological Changes:** Over this six-year span, United transitioned from David Moyes to Louis van Gaal to José Mourinho, each with distinct tactical philosophies and systems. Moyes preferred a pragmatic approach reliant on crossing and long balls. Van Gaal favoured a possession-dominant, controlled style of football, while Mourinho employed a compact defensive structure with direct attacking transitions. These were three distinct ideologies, each requiring specific player profiles to thrive. Moyes inherited an ageing squad accustomed to quick-transition, high-press football under Sir Alex Ferguson, yet was denied his preferred transfer targets. His system required wingers who would hug the touchline and deliver crosses—players such as Gareth Bale would have suited his approach—yet he was not provided with such profiles. Van Gaal’s setup demanded a midfield controller capable of dictating tempo and orchestrating chance creation, but United only acquired a player of that mould (Paul Pogba) after his departure. While Juan Mata could fill that role, he was simultaneously burdened with goalscoring responsibilities alongside Wayne Rooney, who also operated as a makeshift playmaker due to the lack of clinical attackers under Van Gaal. Mourinho’s system historically relied on a dominant, commanding centre-back—akin to John Terry or Javier Zanetti in his previous sides—but he was given relatively inexperienced centre-halves who required both time and coaching to reach that level. His attack required a talisman capable of holding the ball up and being lethal in transition from direct through balls, yet while Marcus Rashford, Anthony Martial, and Romelu Lukaku fit elements of this profile, none sustained the consistent goalscoring output that a Mourinho team demands.

This inability to secure the right profiles to optimise each system explains the recurring pattern observed across the centroid data: stronger sides outclassed United because these managers lacked the player types to fully realise their tactical frameworks, even if those frameworks contained inherent flaws. Lower-ranked teams leveraged home advantage to exploit these shortcomings, reflected in all centroids with an average `opp_final_pos` > 10 containing more away games than home fixtures. Opponents recognised United’s defensive frailties under Moyes and Van Gaal, as well as the inconsistent attacking output under Mourinho, and often

ceded possession in away fixtures, creating hostile atmospheres to disrupt United’s buildup while seeking scoring opportunities through transitions or set pieces.

- **Investments from the Past:** In Pellegrini’s final season (2015/16), when City finished fourth, centroid data indicated dominance in both possession and chance creation, but an inability to convert opportunities. It was evident that additional goalscoring support for Agüero was required. In Guardiola’s first two summer transfer windows, City invested heavily in attack, acquiring talents such as Sané, Gabriel Jesus, and Bernardo Silva. In defence, ball-playing centre-backs were targeted to complement Otamendi and Kompany, resulting in the arrivals of John Stones and Aymeric Laporte. This recruitment strategy ties directly into a point José Mourinho made in his now infamous “Football Heritage” press conference, in which he listed several key City players—“Otamendi, Kevin De Bruyne, Sterling, Silva, Fernandinho, Agüero: these are **investments from the past**.” While Mourinho’s delivery was clearly dramatised for effect, the claim itself is supported by our model data. All of these players were signed under Pellegrini, with some—such as Silva and Agüero—predating him, having been acquired during Mancini’s tenure. Across three different managerial eras, they remained integral to City’s success. This highlights the strategic value of either recruiting players with the versatility to thrive under different tactical systems or appointing managers with similar footballing philosophies. Both Pellegrini and Guardiola implemented possession-heavy systems based on control, patience, and the creation of space, complemented by a clinical finisher to convert chances. Consequently, Kevin De Bruyne, Fernandinho, and Silva flourished under both, while Agüero maintained an exceptional scoring rate across managers. The combination of these “investments from the past” and the tactical continuity between Pellegrini and Guardiola spared City the kind of squad overhauls that destabilised Manchester United, who attempted them three times in six years. Even in Guardiola’s transitional first season (2016/17), when he was still refining his system, City’s retained core ensured enough quality to secure a third-place finish and nearly 80 points. Once Guardiola’s tactical framework fully took hold, the synergy between inherited quality and targeted new signings turned City into an almost unstoppable force.

What Advanced Metrics Reveal: *xG* and *Possession*

Our model covered two distinct periods: an early era (2013–2015) without readily available *xG* and possession data, and a later era where such advanced metrics were integrated into analysis. Some managerial tenures (e.g., Van Gaal and Pellegrini) spanned both, allowing direct comparisons of the additional context these metrics provide.

- **In-depth understanding of Match Dynamics:** In Van Gaal’s 2014/15 season, only limited metrics—such as opponent quality (`opp_final_pos`), venue (`home_0_Away_1`), and, in some cases, season phase (`matchday` ≥ 19)—were available to interpret performance flaws. These offered only a partial view, often requiring speculative recall or reliance on the eye test. By contrast, the 2015/16 data, enriched with *xG* and possession metrics, revealed much deeper insights into Van Gaal’s tactical shortcomings and United’s creative struggles. Despite averaging 60% possession, United consistently produced fewer chances than their opponents, highlighting the predictable and easily contained nature of their buildup play. This explained the low GF values (0.5 and 0.7273), as the side generated too few quality opportunities to regularly test opposing goalkeepers, let alone score.

- **Areas requiring improvement:** Advanced metrics can also be used to identify structural weaknesses within a team. For instance, sides exhibiting high possession values (e.g., United 2015/16) but producing low shot volumes and xG are likely missing a creative attacking midfielder (#8 or 10) capable of delivering incisive passes to break compact defensive structures. Conversely, teams generating high xG while conceding comparable figures and maintaining low possession may lack midfield or defensive players with the ability to control tempo, retain the ball in critical phases, and manage risk in progression. A large positive xG difference that fails to translate into goals often indicates the absence of a clinical striker or the prolonged unavailability of a primary goalscorer (e.g., City 2015/16 without Agüero). Incorporating further advanced metrics—such as big chances created/missed, goalkeeper save percentage, and successful duels or dribbles by midfielders—can refine this analysis by highlighting deficiencies in specific positional units and providing a more granular evaluation of squad depth.

Improvements and Caveats

Ideally, further model refinement would involve incorporating additional features—such as shots on target, big chances created, and saves—to capture match dynamics in greater detail. However, preliminary tests with these variables introduced substantial noise into the model, as reflected in inconsistent silhouette scores and cluster outputs when paired with K-means clustering. Moreover, many of these advanced statistics are not readily accessible in bulk, requiring match-by-match extraction and extended preprocessing time before analysis. Additionally, there is still an element of randomness in the dataset—and consequently the model—as betting odds, while accounting for form, injuries, and other circumstantial factors, do not fully capture them. Future models could incorporate such variables, but must again consider the trade-off between added complexity, potential noise, and the time required to assemble a dataset covering hundreds of matches.

Despite these constraints, the findings of this study provide a strong foundation for future models that integrate more sophisticated variables to enhance performance analysis. This research began with two primary objectives: first, to establish a mathematical and analytical basis for the contrasting performances of Manchester United and Manchester City over a six-year period; and second, to assess the extent to which advanced metrics—particularly xG and possession—can add contextual depth to performance evaluation. Within the limitations of time, resources, and data availability, the model has produced significant insights that extend beyond the conventional “eye test” in both case studies. Looking ahead, these methods could be extended to predictive frameworks capable of not only explaining past performances but also simulating tactical outcomes under varying conditions, offering practical applications for both performance analysis and strategic planning in elite football.